Title: Eliminating the HCV Virus Among People Who Use Drugs

Background

According to the American Society of Addiction Medicine, hepatitis C is the most common chronic blood-borne infection in the US.[1] There are an estimated 3.5 million people in the US that have chronic hepatitis C, and almost 50% are not aware of their infection. [2] Although these numbers are staggering this data excludes several high-risk populations including the incarcerated and homeless. Therefore the number of US residents infected with HCV is probably at least 4.6 million.[2]

The majority of new HCV infections are due to injection drug use.[3] In North America, an estimated 29% of incarcerated persons have HCV.[4] An estimated 76% of patients who are enrolled in opioid treatment programs in the US also have HCV infection. [5] Acute hepatitis C infection can become chronic in 75-85% of cases, [1 6 7] and approximately 20% of HCV patients develop cirrhosis, and 25% of those develop liver failure, liver cancer or both.[8]

Addiction medicine specialists work in a variety of practice settings including primary care, opioid treatment programs, syringe exchanges, addiction centers and safety net programs for the homeless. Since many of these high-risk HCV patients do not see HCPs regularly, addiction medicine healthcare professionals (HCPs) are uniquely poised to address gaps in HCV care for this high-risk population. There are three critical gaps that need to be addressed for the optimal treatment and management of HCV. These include:

1. Addiction medicine HCPs are uniquely poised to address the gap in care for people who inject drugs (PWIDs) and have been infected with the hepatitis C virus (HCV) through screening and linkage to care strategies. Therefore education for this group is imperative.
2. Addiction medicine HCPs require on-going education about current treatment standards for HCV, including those patients on opioid maintenance therapy or who are currently using drugs
3. Addiction medicine HCPs need to build communication skills that will enable successful communication with their patients about the problem of HCV reinfection after cure
4. There are practice barriers unique to the addiction medicine setting that need to be addressed.

Gap 1: Addiction medicine HCPs are uniquely poised to address the gap in care for people who inject drugs (PWIDs) and have been infected with the hepatitis C virus (HCV) through screening and linkage to care strategies. Therefore education for this group is imperative.

HCV is primarily transmitted through exposure to contaminated blood. The majority of new HCV infections are due to injection drug use.[3] According to the American Association for the Study of Liver Diseases (AASLD) and the Infectious Diseases Society of America (IDSA) injection drug use accounts for at least 60% of the acute HCV infections in the US and is the most significant risk for HCV infection.[9] Approximately 80% of people who inject drugs (PWIDs) are infected with HCV. Most of these infections are acquired through sharing injection drug use equipment. The users of non-injected illicit drugs are also at risk due to contaminated injection drug use equipment.[1]

The AASLD and the IDSA guidelines recommend testing all person that at a high-risk for HCV infection. [1 9 10 11] This includes PWID and individuals engaging in intranasal drug use. The guidelines recommend the use of an anti-HCV antibody test followed by a positive confirmation with an HCV RNA PCR test. The guidelines recommend that all HCV positive persons should be linked to a practitioner that can provide a comprehensive management strategy that includes assessing for curative antiviral therapies. This includes patients that continue to use alcohol/and or drugs as well as those patients that are currently receiving medications for opioid use disorder.[1] Screening for HCV based on HCV prevalence has proven benefits for care and treatment including reducing the risks for hepatocellular carcinoma, all-cause mortality as well as the public health benefit for reducing transmission through early treatment, viral clearance, and reduced risk behavior. [9 12 13 14]

The goal for chronic HCV antiviral therapy is to achieve a sustained virological response (SVR). SVR is defined as the continued absence of detectable virus for at least 12-24 weeks after therapy is complete. The current treatment standards recommend that patients be treated in the early stages of HCV infection before the development of severe liver disease or extrahepatic manifestations.[3] The current AASLD and IDSA guidelines recommend treating all patients who have chronic HCV except those with shorter life expectancies that cannot be treated. [10 11] The American Society of Addiction Medicine Policy Statement emphasizes that "State of the art medical treatment for HCV should be accessible and available to all current and former drug users with chronic HCV infection. **Active alcohol and/or drug use should not in itself exclude any person from receiving treatment for their HCV infection.** The outcomes for patients who continue to use alcohol and/or drugs are

comparable to those who do not. The decision to initiate treatment for HCV infection in any patient should be made by the patient and his/her practitioner following careful consideration of the benefits and risks of therapy."[1] Some countries still have restrictions for treatment of PWIDs with DAAs based on the drug and alcohol use criteria. There is a clear need for evidence-based education that validates how these patients can be successfully treated with DAAs.[15] Therefore it is imperative that addiction medicine specialists receive education that allows them to facilitate receive appropriate screening and linkage to care for high-risk HCV patients.

Gap 2: Addiction medicine HCPs require on-going education about current treatment standards for HCV, including those patients on opioid maintenance therapy or who are currently using drugs

An estimated 76% of patients who are enrolled in opioid treatment programs in the US also have HCV infection.[5] These patients need to receive access to addiction treatment services that include pharmacologic treatment.[1] Some practitioners, payers, and policy makers have argued that expensive HCV medication should not be given to people who continue to use alcohol or drugs due to the risk for reinfection.[1] The NIH has stated that "HCV therapy has been successful even when patients have not abstained from continued drug or alcohol use or are on daily methadone."[8] Therefore it is imperative that these patients not be ignored. Additionally, there is data that shows that reinfection rates amongst patients with active alcohol and drug use are low.[16] Therefore the treatment of these individuals will provide the public health benefit of reducing the number of HCV carriers and transmission to others.[1]

The approval of direct-acting antivirals (DAAs) in 2014 has changed the landscape for HCV treatment.[10] These DAAs used in combination have demonstrated greater than 90% efficacy across multiple genotypes even in HCV patients with cirrhosis.[17] [18] [19] [20] [21] [22] [23] [24] The DAAs have had a very positive impact on HCV treatment with SVR rates of more than 90%. Since 2014 there have been several DAAs approved that vary based on genotype targeted and the assessment of the degree of liver cirrhosis. (see table 1) DAA combination products usually consist of two to three DAAs that inhibit the proteins that allow for HCV RNA replication.

Table 1: DAAs currently approved for HCV treatment

DAA	Indications	Chronic HCV Genotype
Daclatasvir[17] in combination with sofosbuvir [25] with our	N5A inhibitor indicated for use with sofosbuvir with or without ribavirin	1 or 3

without ribavirin		
Sofosbuvir[25]	NS5B polymerase inhibitor approved as a component of a combination antiviral treatment regimen for patients without cirrhosis or with compensated cirrhosis.	1-4 without cirrhosis or with compensated cirrhosis
Grazoprevir/elbasvir[22]	Elbasvir is an N5A inhibitor, and grazoprevir is an NS3/4A protease inhibitor indicated for use with ribavirin in certain patient populations	1 or 4 without decompensated cirrhosis
Ombitasvir/paritaprevir/ritonavir[20]	fixed-dose combination of ombitasvir(NS5A inhibitor), paritaprevir, (NS3/4A protease inhibitor), and ritonavir (CYP3A inhibitor) indicated in combination with ribavirin	4 without cirrhosis or with compensated cirrhosis.
Dasabuvir/ombitasvir/paritaprevir/ritonavir[21]	Extended-release combination dasabuvir (non-nucleoside NS5B palm polymerase inhibitor), ombitasvir (NS5A inhibitor), paritaprevir (NS3/4A protease inhibitor), and ritonavir (CYP3A inhibitor) indicated for the treatment of adult patients with chronic HCV	1b without cirrhosis or with compensated cirrhosis 1a without cirrhosis or with compensated cirrhosis for use in combination with ribavirin.
Ledipasvir/sofosbuvir [19]	fixed-dose combination of ledipasvir (NS5A inhibitor), and sofosbuvir (nucleotide analog NS5B polymerase inhibitor) indicated for the treatment of chronic hepatitis C virus in certain populations.	1, 4, 5, 6 Adults with genotype 1,4,5, or 6 infections without cirrhosis or with compensated cirrhosis Adults with genotype 1 infection with decompensated cirrhosis, in combination with ribavirin Adults with genotype 1 or 4 infections who are liver transplant recipients without cirrhosis or with compensated cirrhosis, in combination with ribavirin Pediatric patients 12 years of age and older or weighing at least 35 kg with genotype 1, 4, 5, or 6 without cirrhosis or with compensated cirrhosis.
Sofosbuvir/velpatasvir[18]	fixed-dose combination of sofosbuvir (NS5B polymerase inhibitor), and velpatasvir, (NS5A inhibitor), indicated for the treatment of adult patients with chronic HCV genotype infection	1, 2, 3, 4, 5, or 6 without cirrhosis or with compensated cirrhosis with decompensated cirrhosis for use in combination with ribavirin
Sofosbuvir/velpatasvir/voxilaprevir,[23]	fixed-dose combination of sofosbuvir (NS5B polymerase inhibitor), velpatasvir (HCV NS5A inhibitor), and voxilaprevir, (HCV NS3/4A protease inhibitor), indicated for the treatment of adult patients with chronic HCV infection without cirrhosis or with compensated cirrhosis	1, 2, 3, 4, 5, or 6 infections and have previously been treated with an HCV regimen containing an NS5A inhibitor. genotype 1a or 3 infections and have previously been treated with an HCV regimen containing sofosbuvir without an NS5A inhibitor. An additional benefit of VOSEVI

		over sofosbuvir/velpatasvir was not shown in adults with genotype 1b, 2, 4, 5, or 6 infections previously treated with sofosbuvir without an NS5A inhibitor.
Simprevir[24]	NS3/4A protease inhibitor indicated for the treatment of adults with chronic hepatitis C virus (HCV) infection	in combination with sofosbuvir in patients with HCV genotype 1 without cirrhosis or with compensated cirrhosis in combination with peginterferon alfa (Peg-IFN-alfa) and ribavirin (RBV) in patients with HCV genotype 1 or 4 without cirrhosis or with compensated cirrhosis.
Glecaprevir/pibrentasvir	approved for use in patients with chronic kidney disease disease with or without liver disease.[26]	All genotypes

PWIDs have typically been excluded from clinical trials for new HCV therapies. However, the CO-STAR trial was the first clinical trial that specifically looked at PWIDs receiving opioid substitution therapies. Patients were not excluded if they continued to use illicit drugs. The trial demonstrated that 91% of patients that were treated with grazoprevir/elbasvir in the immediate and deferred groups achieved a combined achieved SVR12 (undetectable HCV RNA at 12 weeks after completing therapy).[15] [27] The trial demonstrated the importance of treating this previously ignored patient population.

Gap 3: Addiction medicine HCPs need to build communication skills that will enable successful communication with their patients about the problem of HCV reinfection after cure

Addiction treatment programs are well positioned to provide and coordinate HCV screening, treatment, and counseling. These counselors need to educate their patients about the potential of HCV reinfection after cure. Since there are many strains of HCV patients can also become re-infected with a new strain or the same strain. There are studies that indicate that reinfection with HCV is possible after achieving an SVR, especially in PWID populations.[28] The fact that some patients get re-infected supports the need for harm reduction services as well as ready access to clean injection supplies. According to Mary Ellen Harrod of the New South Wales Users and AIDS Association, “Reinfection is not about being careless, it is about education and the availability of appropriate healthcare."[15] HCV is easily transmitted through shared drug equipment. Therefore it is imperative that addiction medicine HCPs develop the appropriate communication skills that will

enable them to communicate the risks for HCV infection to their patients. HCPs need to communicate to their patients the importance of practicing safe sex, only visiting licensed tattoo parlors, using clean needles and not coming into contact with blood.

Gap 4: There are practice barriers unique to the addiction medicine setting that need to be addressed.

Addiction medicine HCPs are in the unique position of dealing with high-risk HCV patients on a regular basis. These patients often do not have regular access to other HCPs. Therefore addiction medicine specialists can address the unique needs of this populations. Additional practice barriers that are unique to the addiction medicine setting include the need for:

- *Alcohol counseling and prevention of liver damage*
- *Education on HCV/HIV coinfection*
- *Implementing syringe exchange programs and free condom distribution*

Alcohol counseling and prevention of liver damage

There are several important interventions that can maintain and improve the health of HCV infected PWIDs. The primary focus of treatment is to prevent further liver damage in these patients. Therefore patients who continue to abuse alcohol need to be encouraged to eliminate or significantly reduce alcohol. The American Society of Addiction Medicine Policy emphasizes the use of alcohol-focused screening and brief interventions as outlined by the National Institute of Alcohol Abuse and Alcoholism (NIAAA). These interventions have been demonstrated to reduce the consumption of alcohol and binge drinking in both HCV infected individuals and the general population. Individualized approached to reduce alcohol consumption, and support abstinence should also be encouraged. Additional measures to prevent liver damage include screening for other hepatitis A and B viruses as well as vaccination against these viruses.[1]

Education on HCV/HIV coinfection

HCV and HIV have shared transmission modes, so patients who are living with HIV are also at high risk of infection with HCV.[29 30] Sexual transmission is a particular risk for men who have HIV-infected and have unprotected sex with men.[9] Coinfection with HIV can accelerate the progression of HCV-related liver disease. Therefore the public education of the risks for HIV/HCV coinfection is essential. Access to free testing for patients exposed to both HCV and HIV is important to identify these individuals. The CDC recommends that patients with chronic HCV be

placed on antiretroviral therapy to treat coinfection with HIV. They also recommend pre-exposure prophylaxis for patients that have chronic HCV and have not been exposed to HIV.[1]

Implementing syringe exchange programs and free condom distribution
The incorporation of programs that allow for syringe exchanges and free condom distributions can significantly decrease the frequency of contaminated intravenous drug use injections and associated high-risk unprotected sex.[1]

Gap	Learning Objective	Desired Outcomes
Addiction medicine HCPs are uniquely poised to address the gap in care for people who inject drugs (PWIDs) and have been infected with the hepatitis C virus (HCV) through screening and linkage to care strategies. Therefore education for this group is imperative.	Evaluate the importance of HCV screening and strategies for linkage to care	HCPs will be able to describe the HCV screening process and develop linkage to care models appropriate for use in their unique practice settings, successfully demonstrating ways to improve access to care for their patients
Addiction medicine HCPs require on-going education about current treatment standards for HCV, including those patients on opioid maintenance therapy or who are currently using drugs	Assess individualized patient factors for potential treatment utilizing the latest data on cure rates	Enable HCP's to integrate HCV treatment in with addiction treatment
Addiction medicine HCPs need to build communication skills that will enable successful communication with their patients about the problem of HCV reinfection after cure	Develop counseling strategies for patients that address the problem of reinfection such as appropriate harm reduction	Empower patients to protect themselves from getting reinfected with HCV after cure
There are practice barriers unique to the addiction medicine setting that need to be addressed.	Assess strategies for overcoming practice barriers that are unique to addiction medicine	Develop innovative models of care to address practice barriers

Learning Objectives

1. Evaluate the importance of HCV screening and strategies for linkage to care
2. Assess individualized patient factors for potential treatment utilizing the latest data on cure rates
3. Develop counseling strategies for patients that address the problem of reinfection such as appropriate harm reduction
4. Assess strategies for overcoming practice barriers that are unique to addiction medicine

Agenda

1. The importance of HCV screening and developing strategies for linkage to care (LO1)
    a. The role of addiction medicine HCPs
    b. Screening for high-risk HCV
        i. PWIDs
        ii. Incarcerated people
        iii. Opioid treatment programs
        iv. Homeless
    c. Linking screening to appropriate care programs and practitioners
2. Individualized patient factors for potential treatment utilizing the latest data on cure rates (LO2)
    a. Coinfection with HIV
    b. Genotype
    c. Comorbidities
    d. Liver cirrhosis
    e. High-risk HCV patients
3. Counsel patients on the problem of reinfection and appropriate harm reduction strategies (LO3)
    a. HCV reinfection
    b. Other harm reduction strategies
4. Strategies for overcoming practice barriers unique to addiction medicine (LO4)
    a. Alcohol counseling
    b. Education on HCV/HIV coinfection
    c. Syringe exchange programs and free condom distribution